

Article

Green Synthesis of Activated Carbons from Coconut Coir Dust via Steam Activation for Supercapacitor Electrode Applications

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Abstract

Activated carbons derived from coconut coir dust were synthesized via a two-step process combining carbonization and steam activation for application as electrode materials in supercapacitors. The influence of carbonization temperature (500–700 °C) on the morphological, structural, textural, and electrochemical properties of the resulting activated carbons was systematically investigated. Increasing the carbonization temperature led to a progressive collapse of the cellular structure and formation of a more compact and thermally stable carbon matrix, while the overall morphology remained largely unchanged after steam activation. The steam-activated carbon prepared from the carbonized sample at 700 °C (SA-CCD-7) exhibited the highest specific surface area (889 m² g^{−1}) and a well-developed hierarchical micro–mesoporous structure. Structural analyses confirmed the amorphous nature and an increase in structural disorder after activation, consistent with the enhanced pore development. Electrochemical measurements in 6 M KOH using a three-electrode system revealed that the SA-CCD-7 displayed a typical electric double-layer capacitor (EDLC) behavior, delivering the highest specific capacitance of 86 F g^{−1} at 1 A g^{−1} and retaining 81% of its initial capacitance at 20 A g^{−1}, demonstrating excellent rate capability. The symmetric coin-cell supercapacitor device assembled with SA-CCD-7 as the electrodes achieved an energy density of 0.9–1.2 Wh kg^{−1} and a power density of 50–2500 W kg^{−1}, along with remarkable cycling stability over 10,000 cycles with negligible capacitance loss. These findings highlight steam activation of coconut coir dust as a simple, scalable, and eco-friendly approach for producing biomass-derived carbon electrodes for sustainable energy storage applications.

Keywords: activated carbons; steam activation; coconut coir dust; supercapacitors



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1. Introduction

In recent years, the rapid growth of portable electronic devices, electric vehicles, and renewable energy systems has created an urgent demand for advanced energy-storage technologies that can deliver both high power and long-term reliability [1]. Batteries and supercapacitors represent the two principal classes of electrochemical energy-storage devices [2]. Batteries, such as lithium-ion and sodium-ion systems, offer high energy densities

but suffer from relatively slow charge–discharge kinetics and limited cycle life [3,4]. In contrast, supercapacitors bridge the gap between conventional dielectric capacitors and batteries by combining the high power density and rapid charge–discharge capability of capacitors with the moderate energy density of batteries. These distinctive features make supercapacitors highly attractive for applications such as regenerative braking, power backup units, grid stabilization, and hybrid energy modules [5]. However, the energy density of supercapacitors remains significantly lower than that of batteries, which constrains their broader adoption. Enhancing their energy-storage capability depends critically on the development of advanced electrode materials, which play a decisive role in determining both capacitance and rate performance. The electrochemical behavior of supercapacitors is influenced by several intrinsic characteristics of the electrode materials, including specific surface area, pore architecture, surface functionality, and electrical conductivity [6–8]. Therefore, the rational design and controlled synthesis of electrode materials with optimized textural and structural properties have become central to the advancement of next-generation supercapacitors [9].

Activated carbons (ACs) have been the most widely used and commercially successful electrode material owing to their high specific surface area, tunable pore structure, and cost-effectiveness [10,11]. Recently, biomass-derived ACs have emerged as sustainable alternatives, offering the dual benefits of waste utilization and providing high-performance materials for energy storage applications [12–14]. The unique composition of biomass, comprising cellulose, hemicellulose, and lignin, facilitates the formation of a well-connected carbon framework with tunable porosity upon thermal treatment and activation. Various agricultural and agro-industrial residues, such as coconut shells [15], cashew nutshells [16], palm shells [17], corncobs [18], rice husks [19], and fruit peels [20–22], have been successfully converted into ACs for supercapacitor electrodes. The valorization of such locally abundant and underutilized biomass resources not only alleviates waste management problems but also supports the principles of circular economy and green chemistry [23].

Among biomass precursors, coconut coir dust, a lignocellulosic byproduct of the coconut industry, represents an underutilized yet promising carbon source for producing ACs. Unlike dense and woody coconut shells, coconut coir dust possesses a looser and more fibrous morphology, higher volatile matter, and lower ash content, which collectively promote the development of a more open and interconnected pore network during activation. These intrinsic characteristics make coconut coir dust particularly attractive for supercapacitor electrode applications. However, while coconut coir dust-derived ACs have been extensively investigated as adsorbents for pollutant removal [24–26], their potential in electrochemical energy storage remains unexplored. Addressing this knowledge gap is therefore essential to expand the applicability of coconut coir dust-derived ACs beyond environmental remediation and toward sustainable energy-storage technologies.

ACs can generally be synthesized through physical or chemical activation routes [27]. Among them, physical activation using steam is an environmentally benign method to generate porous carbon without the use of corrosive or hazardous activating agents. In this process, carbonized precursors are exposed to steam at elevated temperatures, where gasification selectively removes carbon atoms from less ordered regions, thereby generating interconnected micropores and mesopores [28]. Compared to chemical activation, steam activation offers several advantages: it minimizes chemical waste, allows better control of pore development, and promotes graphitic ordering due to the high activation temperature. Moreover, the steam activation process can be readily scaled up for industrial production, making it highly attractive for the sustainable manufacturing of ACs [29–31]. Despite these advantages, research on the steam activation of coconut coir dust remains limited, particularly concerning its structure–property–performance relationships in supercapacitor

applications. A systematic investigation into how processing parameters affect the resulting structure and electrochemical behavior is therefore essential to fully realize its potential as a high-performance electrode material.

In this work, coconut coir dust was employed as a renewable precursor for the preparation of ACs via a two-step process comprising carbonization and steam activation. The precursor was first carbonized at different temperatures (500, 600, and 700 °C) and subsequently subjected to steam activation at 900 °C to develop porosity and enhance surface area. The resulting ACs were comprehensively characterized using physicochemical and electrochemical techniques to elucidate the relationships among microstructure, surface chemistry, and capacitive performance. This study provides fundamental insights into the role of carbonization temperature in optimizing the pore architecture and charge storage behavior of ACs derived from coconut coir dust via steam activation.

2. Materials and Methods

2.1. Materials

The coconut coir dust was collected from the local market in Thailand. It was passed through a 150-mesh screen to remove coarse coconut fibers and obtain fine powders. Potassium hydroxide (KOH), isopropanol (C₃H₈O, 99.8%), and *N*-methyl-2-pyrrolidone (C₅H₉NO, NMP, 99.5%) were purchased from LabScan Co., Ltd. (Bangkok, Thailand). Polyvinylidene fluoride (PVDF, molecular weight ~ 534,000 g mol⁻¹) and Nafion® DE 521 solution (5 wt% in a mixture of lower aliphatic alcohols and water) were supplied by Sigma-Aldrich (St. Louis, MO, USA). Potassium bromide (KBr, >99.0%) was acquired from Thermo Scientific Chemicals (Haverhill, MA, USA). Acetylene black and graphite sheets were supplied from AME Energy Co., Ltd. (Shandong, China), while a cellulose separator (30 µm-thick) was sourced from the Nippon Kodoshi Corporation (Kochi, Japan). Argon gas (industrial grade, 99.95%) was purchased from Linde Public Company Limited (Bangkok, Thailand). All chemicals were of analytical grade and used as received without further purification.

2.2. Preparation of Steam-Activated Carbons from Coconut Coir Dust

Coconut coir dust was first subjected to carbonization at 500, 600, and 700 °C for 1 h in a Vecstar tube furnace (Chesterfield, UK) under an argon atmosphere, with a heating rate of 5 °C min⁻¹. After cooling to room temperature, the resulting carbonized products were gently ground using a mortar and pestle to obtain fine powders. The samples carbonized at 500, 600, and 700 °C were designated as CCD-5, CCD-6, and CCD-7, respectively. Subsequently, each CCD sample was activated with steam in the same tube furnace at 900 °C for 2 h under a continuous argon flow (0.2 L min⁻¹) using the same heating rate of 5 °C min⁻¹. After natural cooling to room temperature, the steam-activated products were thoroughly ground to obtain fine carbon powders. The resulting steam-activated carbons derived from CCD-5, CCD-6, and CCD-7 were labeled as SA-CCD-5, SA-CCD-6, and SA-CCD-7, respectively. A schematic illustration of the overall preparation process is shown in Figure 1.

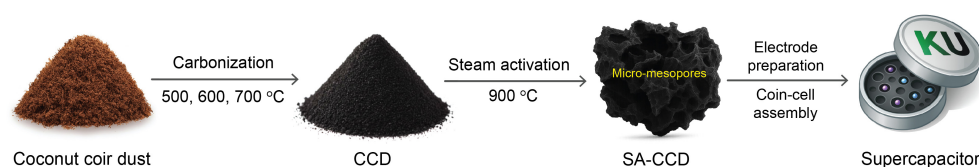


Figure 1. Schematic illustration of the carbonization and steam activation steps used to prepare SA-CCD in this work.

2.3. Characterization

The surface morphology of the samples was characterized using a Hitachi SU3500 scanning electron microscope (SEM, Tokyo, Japan) operated at an accelerating voltage of 5 kV. Energy-dispersive X-ray spectroscopy (EDS) was carried out with an Oxford Instrument detector (High Wycombe, UK) equipped with the SEM system. Textural properties, including specific surface area and pore characteristics, were determined from N₂ adsorption–desorption isotherms at −196 °C using a Micromeritics 3Flex analyzer (Norcross, GA, USA), after degassing at 150 °C for 12 h with a Smart VacPrep system (Norcross, GA, USA). Crystalline structures were analyzed by X-ray diffraction (XRD) using a PANalytical Empyrean diffractometer (Worcestershire, UK) equipped with Cu K α radiation ($\lambda = 1.5406 \text{ \AA}$), operated at 40 kV and 30 mA. Elemental composition was examined using a Horiba XGT-5000 X-ray fluorescence (XRF) spectrometer (Kyoto, Japan). Raman spectra were recorded on a Thermo Fisher Scientific DXR SmartRaman spectrometer (Waltham, MA, USA) employing a 532 nm excitation laser. Chemical functional groups were analyzed using a Bruker Alpha-E Fourier transform infrared (FTIR) spectrometer (Billerica, MA, USA) in transmission mode. The samples were mixed with KBr and pressed into a transparent pellet prior to measurement.

2.4. Electrochemical Measurements

For the electrochemical evaluation in the three-electrode system, 5 mg of SA-CCD powder was ultrasonically dispersed for 1 h in a mixed solvent containing 475 μL of deionized water, 475 μL of isopropanol, and 50 μL of Nafion[®] DE 521 to obtain a uniform suspension. A glassy carbon (GC) electrode (3 mm diameter, ALS Co., Ltd., Tokyo, Japan) was carefully polished using 0.1 μm diamond and 0.05 μm alumina slurries, followed by rinsing with deionized water and air drying. A 3 μL aliquot of the prepared suspension was drop-cast onto the GC electrode surface and dried for 3 h under ambient conditions, resulting in a mass loading of 0.015 mg.

All electrochemical measurements were conducted at room temperature in 6 M KOH electrolyte using a three-electrode system. A platinum wire (ALS Co., Ltd., Tokyo, Japan) and Hg/HgO (in 1 M NaOH; ALS Co., Ltd., Tokyo, Japan) served as the counter and reference electrodes, respectively, while the SA-CCD-modified GC electrodes functioned as the working electrodes. Cyclic voltammetry (CV) and galvanostatic charge–discharge (GCD) tests were performed using a Biologic VSP potentiostat/galvanostat controlled by EC-Lab software (version 10.33).

The specific capacitance (C_s) of SA-CCD electrodes was calculated from the GCD curves according to the following equation:

$$C_s = \frac{2I}{m(V_f - V_i)^2} \int_{V_i}^{V_f} V dt \quad (1)$$

where I is the applied current, m is the mass of the active material, $\int V dt$ is the integrated area under the discharge curves, and V_i and V_f are the initial and final potentials, respectively.

For the two-electrode system, the best-performing SA-CCD sample was used as the electrodes to assemble a symmetric coin-cell supercapacitor (CR2032 type). The electrode slurry was prepared by mixing SA-CCD, acetylene black, and PVDF binder in a weight ratio of 7:1:2, with NMP as the solvent. The homogeneous slurry was coated onto a graphite sheet with an automatic film applicator (KV-AFA-L280, Zhuhai Kaivo Optoelectronic Technology Co., Ltd., Zhuhai, China), followed by drying at 80 °C for 12 h. The electrodes and cellulose separators were cut into disks of 16 mm and 19 mm in diameter, respectively, using a CP60 die cutter punching machine (Xiamen Tmax Battery Equipments Limited, Xiamen, China). Each electrode contained approximately 4 mg of active material per side.

Two identical electrodes were assembled face-to-face with a separator saturated in 6 M KOH electrolyte. The cells were sealed using an MSK-110 hydraulic crimping machine (Xiamen Tmax Battery Equipments Limited, Xiamen, China).

The specific capacitance of the cell (C_{cell}) was determined from the GCD curves using the following equation:

$$C_{\text{cell}} = \frac{I \times \Delta t}{m \times \Delta V} \quad (2)$$

where I is the applied current, Δt is the discharge time, m is the total mass of active material on both electrodes, and ΔV is the potential window.

The corresponding energy density (E , Wh kg^{−1}) and power density (P , W kg^{−1}) of the cell were calculated using the following equations:

$$\text{Energy density : } E = \frac{C_{\text{cell}} \times (\Delta V^2)}{2 \times 3.6} \quad (3)$$

$$\text{Power density : } P = \frac{E \times 3600}{\Delta t} \quad (4)$$

3. Results and Discussion

3.1. Morphology

The surface morphologies of the carbonized and activated samples are shown in Figure 2. At the lowest carbonization temperature (500 °C), the CCD-5 retained a relatively well-preserved cellular framework with clearly defined pore channels, reflecting the intrinsic microstructure of the original coconut coir dust (Figure S1). As the carbonization temperature increased to 600 °C and 700 °C, progressive structural collapse and densification of the cell walls were observed. This morphological evolution is attributed to the enhanced thermal decomposition of hemicellulose and cellulose, accompanied by the release of volatile matter and softening of the carbon matrix during pyrolysis. Consequently, the surface became smoother and more compact, with a noticeable reduction in macropore openness. After steam activation at 900 °C, the overall morphology of the SA-CCD samples remained largely similar to that of their carbonized counterparts, indicating that the activation process did not significantly alter the macroscopic cellular framework. The preservation of the skeleton structure suggests that steam activation primarily promoted internal pore development, particularly the formation of micro- and mesopores within the carbon matrix, rather than the extensive external etching.

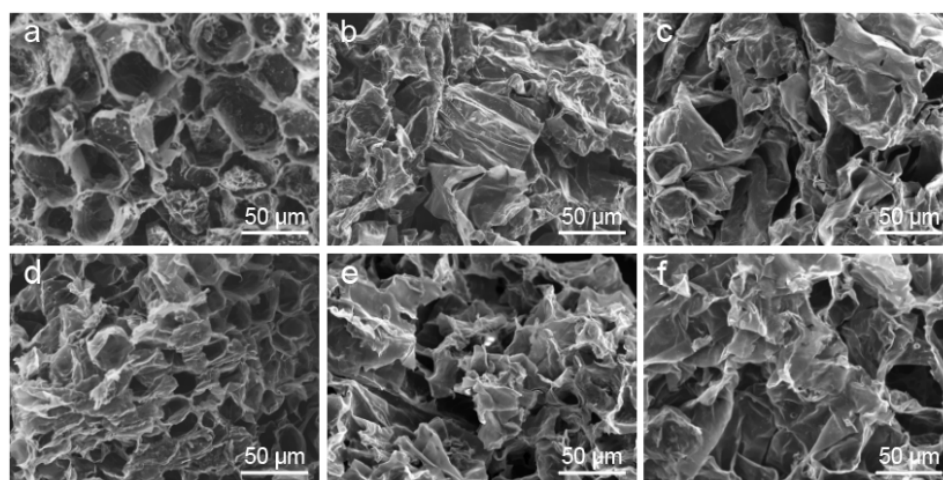


Figure 2. SEM images of (a) CCD-5, (b) CCD-6, (c) CCD-7, (d) SA-CCD-5, (e) SA-CCD-6, and (f) SA-CCD-7.

3.2. Surface Area and Porosity

The textural characteristics of the CCD and SA-CCD samples were evaluated by N₂ adsorption–desorption measurements, as shown in Figure 3a. The specific surface area (S_{BET}) was determined by the Brunauer–Emmett–Teller (BET) method. The isotherm of CCD-5 showed very low N₂ adsorption with an S_{BET} of only 7 m² g^{−1}, indicating that carbonization at 500 °C was insufficient to develop a porous structure. This observation is consistent with the SEM results, which showed an intact cellular morphology. When the carbonization temperature increased to 600 and 700 °C, a substantial rise in N₂ uptake was observed for CCD-6 and CCD-7, both exhibiting a type I isotherm and higher S_{BET} values of 140 and 203 m² g^{−1}, respectively. This enhancement can be attributed to the progressive decomposition of lignocellulosic components and the release of volatile species, which generate primary micropores within the carbon matrix. According to the *t*-plot analysis, the micropore surface area (S_{micro}) accounted for approximately 79% and 85% of S_{BET} for CCD-6 and CCD-7, respectively, confirming that micropores dominate the porosity of the carbonized samples. After steam activation, all samples showed a pronounced increase in N₂ adsorption, particularly at both low ($P/P_0 < 0.01$) and intermediate ($P/P_0 = 0.45–0.99$) relative pressures, indicating the formation of a hierarchical micro–mesoporous structure. The isotherm type evolved from type I to type IV, characteristic of mesoporous materials with capillary condensation in the medium-pressure region. The S_{BET} values of SA-CCD-5, SA-CCD-6, and SA-CCD-7 significantly increased to 779, 802, and 889 m² g^{−1}, respectively, accompanied by a corresponding rise in total pore volume (V_{total}) from 0.536 to 0.715 cm³ g^{−1}. This significant enhancement in surface area and pore volume confirms that steam activation effectively develops internal porosity within the carbon framework.

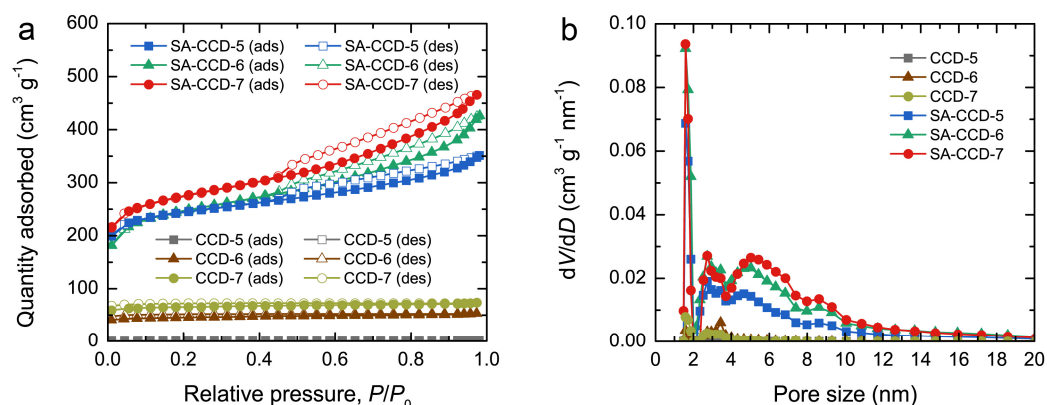


Figure 3. (a) N₂ adsorption–desorption isotherms and (b) DFT pore-size distribution profiles of CCD and SA-CCD samples.

The S_{BET} of SA-CCD-7 (889 m² g^{−1}) is comparable to that of ACs synthesized from various coconut-based precursors via physical activations but remains lower than the values (>1000 m² g^{−1}) typically achieved by chemical activation (Table S1). This difference arises from the use of steam as a mild physical activating agent, which creates porosity through controlled gasification rather than aggressive chemical etching, thereby preserving the structural integrity of the carbon matrix [31]. Although chemical activation can produce higher surface areas, it often leads to excessive burn-off, pore collapse, and the use of corrosive chemicals [32]. Further enhancement of the surface area and more precise control over the pore structure could be achieved by optimizing activation parameters such as temperature, duration, and steam flow rate.

The pore-size distribution curves derived from the density functional theory (DFT) (Figure 3b) further confirm the coexistence of micropores (<2 nm) and small mesopores (2–10 nm) in all activated samples. The mesopore contribution increased progressively

with carbonization temperature, suggesting pore widening and enhanced connectivity during activation. Quantitatively, the S_{micro} increased from $253 \text{ m}^2 \text{ g}^{-1}$ (SA-CCD-5) to $358 \text{ m}^2 \text{ g}^{-1}$ (SA-CCD-7), while the mesopore volume (V_{meso}) expanded from 0.268 to $0.446 \text{ cm}^3 \text{ g}^{-1}$. Although the proportion of S_{micro} relative to S_{BET} decreased to 57–60% after activation, these micropores continued to contribute significantly (50–60%) to the total pore volume, ensuring an optimal balance between ion adsorption sites and electrolyte transport pathways. Textural parameters obtained from the N_2 adsorption–desorption isotherm analysis of CCD and SA-CCD samples are summarized in Table 1.

Table 1. Textural parameters obtained from the N_2 adsorption–desorption isotherm analysis of CCD and SA-CCD samples.

Sample	S_{BET} ($\text{m}^2 \text{ g}^{-1}$)	S_{micro} ($\text{m}^2 \text{ g}^{-1}$)	S_{meso} ($\text{m}^2 \text{ g}^{-1}$)	V_{total} ($\text{cm}^3 \text{ g}^{-1}$)	V_{micro} ($\text{cm}^3 \text{ g}^{-1}$)	V_{meso} ($\text{cm}^3 \text{ g}^{-1}$)
CCD-5	7	0	7	0.012	0	0.012
CCD-6	140	111	29	0.082	0.058	0.024
CCD-7	203	172	31	0.113	0.087	0.026
SA-CCD-5	779	526	253	0.536	0.268	0.268
SA-CCD-6	812	507	305	0.644	0.232	0.412
SA-CCD-7	889	531	358	0.715	0.269	0.446

The increasing S_{BET} and V_{total} with higher carbonization temperature (SA-CCD-7 > SA-CCD-6 > SA-CCD-5) indicate that CCD-7 possessed a more stable and ordered carbon skeleton. Such a structure resisted excessive gasification during steam activation and promotes the selective etching of amorphous regions, leading to a well-developed micro–mesoporous network. In contrast, CCD-5 and CCD-6, which contained more amorphous and oxygenated species, were more reactive toward steam, leading to partial over-etching and limited surface development. Overall, SA-CCD-7 exhibited the most optimized hierarchical porosity with the highest S_{BET} and V_{total} , which are expected to enhance electrolyte accessibility and charge storage capability in supercapacitor applications.

3.3. Structural Properties

The structural characteristics of the CCD and SA-CCD samples were examined by XRD and Raman spectroscopy, as shown in Figure 4a,b, respectively. The XRD patterns of all samples exhibited a broad diffraction peak centered around 23° , accompanied by a weaker hump near 43° , corresponding to the (002) and (101) planes of carbon, respectively. These diffuse features indicate that both CCD and SA-CCD consisted of disordered graphitic crystallites embedded within an amorphous carbon matrix, reflecting a low degree of graphitization [33]. With increasing carbonization temperature, the (002) diffraction peak of the CCD-7 became slightly sharper and more defined, suggesting the gradual development of short-range graphitic ordering and partial stacking of carbon layers. After steam activation, the XRD patterns of the SA-CCD samples exhibited a slight increase in low-angle scattering intensity, suggesting the formation of porous structures through selective gasification of less ordered regions during activation [34]. The (002) peaks of the SA-CCD samples became broader than those of their carbonized counterparts, particularly for SA-CCD-7, indicating that the steam activation process introduced additional structural disorder and disrupted existing graphitic domains [33,34]. In addition, weak diffraction peaks were detected in all samples, which can be attributed to residual inorganic species naturally present in the coconut coir dust. To further identify these inorganic residues, both EDS and XRF analyses were performed. The results consistently revealed the presence of silicon (Si), calcium (Ca), and potassium (K), as the major inorganic constituents, with iron (Fe) and zinc (Zn) appearing in smaller amounts (Figures S2 and S3, and Table S2).

These elements originate from the inherent mineral components of coconut coir dust and may remain partially embedded within the carbon matrix after carbonization and steam activation. Although present in small amounts, such residues are likely responsible for the minor crystalline peaks observed in the XRD patterns.

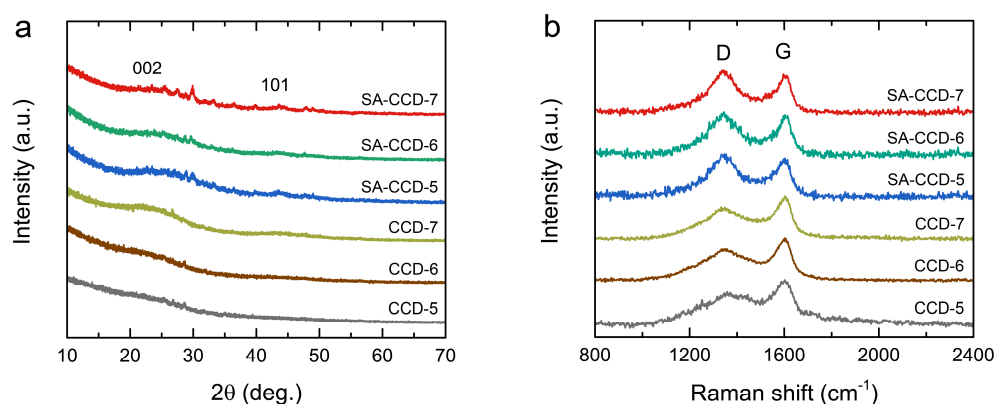


Figure 4. (a) XRD patterns and (b) Raman spectra of the CCD and SA-CCD samples.

The Raman spectra of the CCD and SA-CCD samples (Figure 4b) further support the XRD findings. Two characteristic bands were observed at around 1350 cm^{-1} (D band) and 1595 cm^{-1} (G band), corresponding to the A_{1g} breathing mode of disordered carbon and the E_{2g} vibration mode of sp^2 -hybridized carbon atoms in graphitic domains, respectively [35]. The intensity ratio of D to G bands (I_D/I_G) serves as an indicator of the degree of disorder within the carbon framework [36]. The I_D/I_G ratio increased from 0.63 (CCD-5) to 0.74 (CCD-6) and 0.73 (CCD-7), indicating a gradual rise in structural disorder and defect density with increasing carbonization temperature. This trend is associated with the thermal decomposition of volatile components and the formation of defect sites during carbon-layer rearrangement. After steam activation, the I_D/I_G ratios further increased to 1.17, 1.08, and 1.09 for SA-CCD-5, SA-CCD-6, and SA-CCD-7, respectively. This result confirms that steam activation introduces additional lattice imperfections and defect sites as pores are generated. Overall, the XRD and Raman analyses consistently demonstrate that higher carbonization temperatures and steam activation promote the development of a more disordered and porous carbon structure.

3.4. Chemical Functionality

The FTIR spectra of the raw coconut coir dust, CCD, and SA-CCD are shown in Figure 5. The raw coconut coir dust showed several absorption bands typical of lignocellulosic biomass [37]. A broad peak at 3340 cm^{-1} corresponds to the O–H stretching vibration of hydroxyl groups in cellulose, hemicellulose, and lignin. The peaks at 2937 cm^{-1} and 2854 cm^{-1} are assigned to C–H stretching of aliphatic $-\text{CH}_2$ and $-\text{CH}_3$ groups, respectively. A strong absorption band at 1735 cm^{-1} is attributed to C=O stretching of carbonyl and carboxylic ester groups, whereas the bands at 1613 cm^{-1} and 1522 cm^{-1} correspond to C=C stretching in aromatic rings of lignin. Additional features observed at 1430 and 1037 cm^{-1} are related to C–H and C–O stretching vibrations of alcohol, ether, and phenolic groups [25,38]. These observations confirm the presence of abundant oxygen-containing functional groups in the raw precursor.

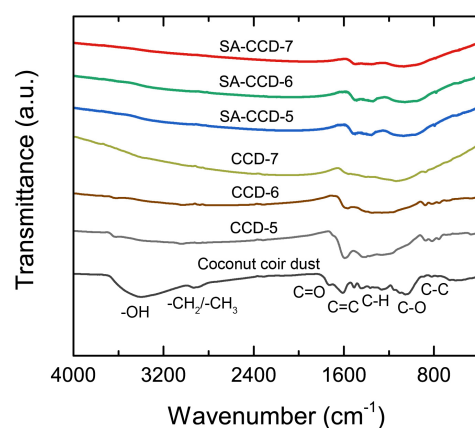


Figure 5. FTIR spectra of raw coconut coir dust, CCD, and SA-CCD samples.

After carbonization (CCD-5, CCD-6, and CCD-7), most of the characteristic bands associated with cellulose and hemicellulose markedly diminished or disappeared, indicating extensive thermal decomposition and removal of volatile components [39]. The weakening or disappearance of the O–H ($\sim 3340\text{ cm}^{-1}$) and C=O ($\sim 1738\text{ cm}^{-1}$) bands demonstrates dehydration and decarbonylation/decarboxylation reactions during pyrolysis. The remaining weak bands around $1580\text{--}1600\text{ cm}^{-1}$ (C=C stretching) and 875 cm^{-1} (C–C bending) suggest the formation of aromatic carbon frameworks. Increasing the carbonization temperature resulted in a progressive loss of oxygenated groups, consistent with enhanced carbonization and aromatization of the carbon matrix. In the SA-CCD samples, the overall intensity of the oxygen-containing groups further decreased, although weak peaks around 1600 cm^{-1} and 1440 cm^{-1} , remained observable. The disappearance of oxygen-containing functional groups (e.g., O–H, C–O, and C=O) indicates efficient removal of residual volatiles and decomposition of remaining organic species during steam activation at $900\text{ }^{\circ}\text{C}$ [40]. The persistence of minor C–O and C=C signals implies that a small fraction of surface oxygen functionalities was retained. Overall, the FTIR results confirm the progressive transformation of the lignocellulosic precursor into carbonaceous structures through carbonization and steam activation at high temperatures.

3.5. Electrochemical Performance

Electrochemical measurements were performed in 6 M KOH using a three-electrode system to evaluate the capacitive behavior of SA-CCD samples. As shown in Figure 6a, the CV curves of SA-CCD-5, SA-CCD-6, and SA-CCD-7, recorded at a scan rate of 50 mV s^{-1} , exhibited a quasi-rectangular shape characteristic of EDLC behavior [41]. The current response followed the order: SA-CCD-7 > SA-CCD-6 > SA-CCD-5. This finding indicates that SA-CCD-7 possessed the highest capacitive performance, primarily attributed to its larger specific surface area and well-developed hierarchical porosity. The rate capability was further evaluated by varying the scan rates from 10 to 100 mV s^{-1} (Figures 6b and S4). The CV curves of all samples retained their quasi-rectangular shape across all scan rates with minimal distortion, confirming excellent ion transport kinetics and low internal resistance [42]. This stable capacitive behavior across a wide range of scan rates suggests that the porous structure produced by steam activation provided efficient ion diffusion pathways and rapid charge propagation throughout the electrode.

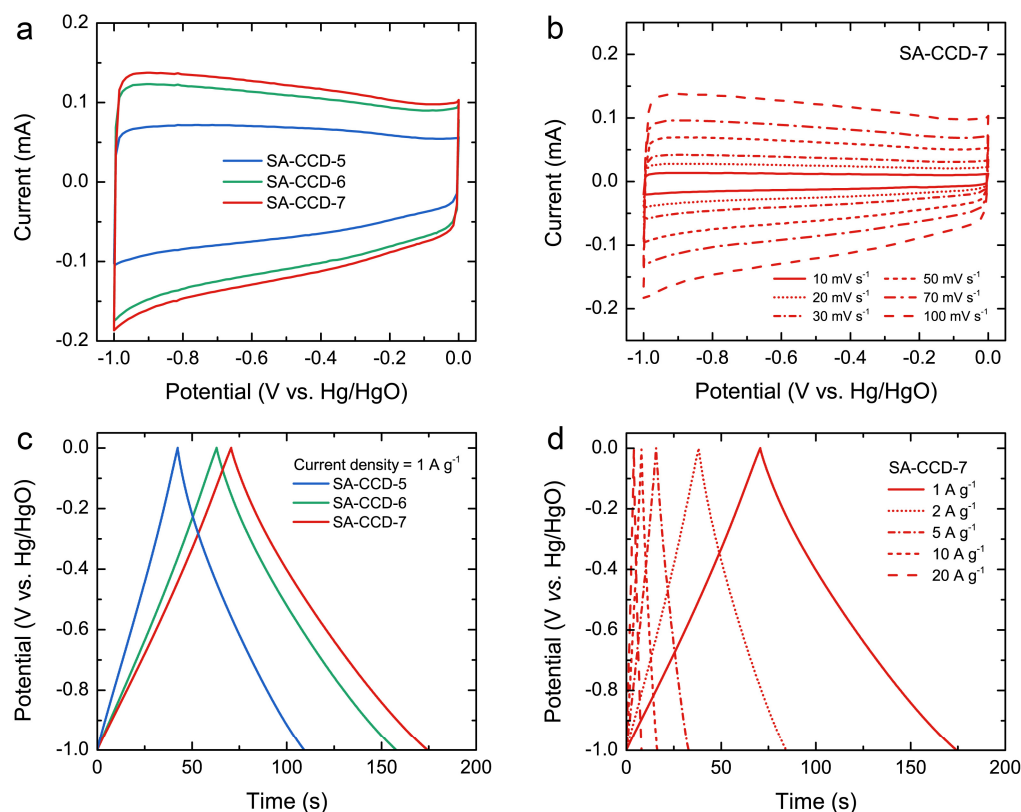


Figure 6. Electrochemical measurements in 6 M KOH electrolyte using a three-electrode system: (a) Comparative CV curves at a scan rate of 50 mV s^{-1} of all SA-CCD samples. (b) CV curves of the SA-CCD-7 at various scan rates varying from 10 to 100 mV s^{-1} . (c) Comparative GCD curves at a current density of 1 A g^{-1} of all SA-CCD samples. (d) GCD curves of SA-CCD-7 at various current densities ranging from 1 to 20 A g^{-1} .

The GCD curves recorded at 1 A g^{-1} (Figure 6c) displayed nearly symmetric triangular shapes for all samples, further validating their dominant EDLC behavior. The discharge time followed the same order as the current response observed in the CV results ($\text{SA-CCD-7} > \text{SA-CCD-6} > \text{SA-CCD-5}$), consistent with the increase in surface area and pore volume. The longer discharge period of SA-CCD-7 indicates its greater charge storage capability. When the current density increased from 1 to 20 A g^{-1} (Figures 6d and S5), all samples maintained linear and symmetric GCD curves with only a moderate decrease in discharge time. The absence of a significant IR drop even at high current densities highlights the good electrical conductivity and robust structural integrity of the steam-activated carbons.

The specific capacitance (C_s) values of SA-CCD-5, SA-CCD-6, and SA-CCD-7, calculated from the GCD curves, were 54 , 78 , and 86 F g^{-1} , respectively, at 1 A g^{-1} . The progressive increase in C_s with carbonization temperature correlates well with the increase in specific surface area, confirming that electrochemical performance is strongly governed by the accessible surface area and pore architecture. The hierarchical micro–mesoporous structure plays a key role: micropores provide abundant adsorption sites for charge accumulation, while mesopores act as ion-buffering reservoirs that promote electrolyte diffusion and reduce ion-transport resistance [43–45]. As the current density increased from 1 to 20 A g^{-1} (Figure 7a), all samples exhibited a gradual decrease in C_s due to the limited time available for electrolyte ions to penetrate deep into the microporous regions during fast charge–discharge processes. Despite this reduction, the capacitance retention remained relatively high, approximately 79%, 79%, and 81% for SA-CCD-5, SA-CCD-6, and SA-CCD-7, respectively, demonstrating excellent rate capability. The superior retention of SA-CCD-7

further highlights the advantage of its interconnected pore network, which effectively balances high surface area with rapid ion diffusion pathways. The C_s value of SA-CCD-7 obtained in this work is comparable to that of biomass-derived ACs prepared by physical activation (steam or CO_2) reported in previous studies (Table 2) [46–52].

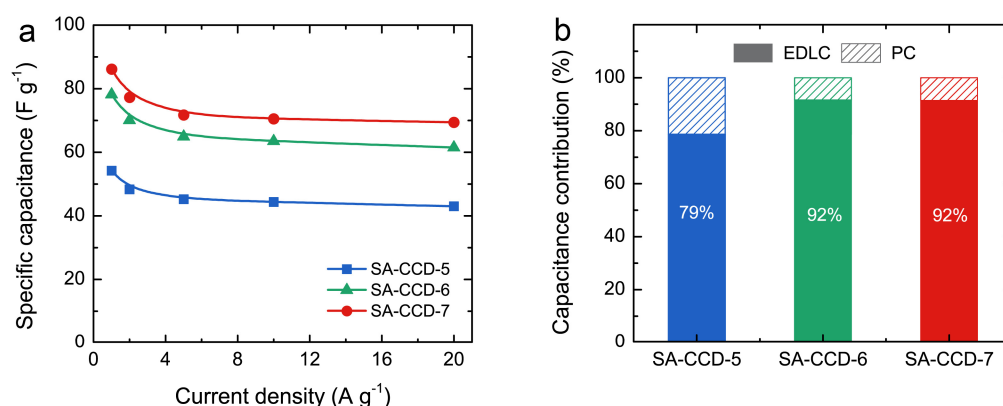


Figure 7. (a) Specific capacitance (C_s) versus current density and (b) capacitance contribution derived from the Trasatti method of SA-CCD samples.

Table 2. Comparison of specific capacitance (C_s) and specific surface area (S_{BET}) of SA-CCD-7 and other biomass-derived AC electrodes prepared by physical activation, measured in a three-electrode system at 1 A g^{-1} .

Sample	Source	Activating Agent	Activation Condition	S_{BET} ($\text{m}^2 \text{ g}^{-1}$)	C_s (F g^{-1})	Electrolyte	Ref.
CAC	Cattail	CO_2	850°C , 2 h	441	110	6 M KOH	[46]
CO_2 activated	Date palm frond	CO_2	900°C , 90 min	604	57	1 M H_2SO_4	[47]
PA-3	Spent coffee grounds	Steam	800°C , 60 min	981	72	1 M Na_2SO_4	[48]
AC-P	Oil palm kernel shell	Steam	500°C , 4 h	727	94	1 M KOH	[49]
AC-S	Oak	Steam	800°C , 40 min	581	105	6 M KOH	[50]
p-KF-9A	Kenaf	Steam	900°C , 30 min	1375	85	6 M KOH	[51]
CSAC	Coconut shell	-	-	780	94	6 M KOH	[52]
SA-CCD-7	Coconut coir	Steam	900°C , 2 h	889	86	6 M KOH	This work

Furthermore, the Trasatti analysis was performed to elucidate the relative contributions of the surface-controlled (EDLC) and diffusion-controlled (pseudocapacitive, PC) processes to the overall charge storage behavior [53]. As shown in Figure 7b, SA-CCD-5 exhibited an EDLC contribution of 79%, while SA-CCD-6 and SA-CCD-7 showed higher values of approximately 92%, indicating that charge storage is predominantly governed by electric double-layer formation. The slightly higher PC contribution in SA-CCD-5 can be attributed to its higher content of oxygen-containing surface functional groups, which facilitate reversible redox reactions [54–56]. In contrast, SA-CCD-6 and SA-CCD-7, possessing a more graphitized and hierarchically porous carbon structure, favor efficient ion adsorption and rapid charge transfer through a purely EDLC mechanism. These results confirm that higher carbonization temperature enhances electrical conductivity and structural stability, while reducing the oxygen-containing surface functional groups, thereby leading to improved electrochemical performance dominated by EDLC behavior.

To evaluate the practical applicability of SA-CCD-7 as an electrode material for supercapacitor devices, a symmetric coin-cell was assembled in a two-electrode configuration. As shown in Figure 8a, the CV curves recorded at scan rates from 10 and 100 mV s^{-1} retained a nearly rectangular shape, confirming the predominant EDLC behavior. The absence of noticeable distortion at higher scan rates indicates excellent rate capability and capacitive performance, reflecting efficient ion transport and low internal resistance within

the electrode material. The GCD curves at current densities from 0.1 to 2 A g⁻¹ exhibited symmetric and linear triangular profiles with negligible *IR* drop, further verifying ideal capacitive behavior and high reversibility of charge–discharge processes (Figure 8b). The specific capacitance of the device (*C*_{cell}), calculated from the GCD curves (Figure 8c), decreased gradually with increasing current density, due to limited ion diffusion at high rates. The device retained a capacitance of 5.4 F g⁻¹ at 2 A g⁻¹, corresponding to 68% of its initial value (8.0 F g⁻¹ at 0.1 A g⁻¹), demonstrating good rate performance and stable charge storage capability. The Ragone plot (Figure 8d) shows that the device delivered an energy density of 0.9–1.2 Wh kg⁻¹ and a power density of 50–2500 W kg⁻¹. The gradual decrease in energy density with increasing power density reflects the typical power–energy trade-off observed in electrochemical supercapacitors, where higher current densities shorten discharge times and increase resistive losses. However, the device maintained a relatively high energy density even at elevated power density. These values are comparable to or higher than those of other biomass-derived ACs reported in the literature [16,57–63]. The inset in Figure 8c further demonstrates the ability of two assembled coin-cell devices connected in series to illuminate a green light-emitting diode (LED), visually confirming the practical energy-storage capability of the fabricated supercapacitor devices.

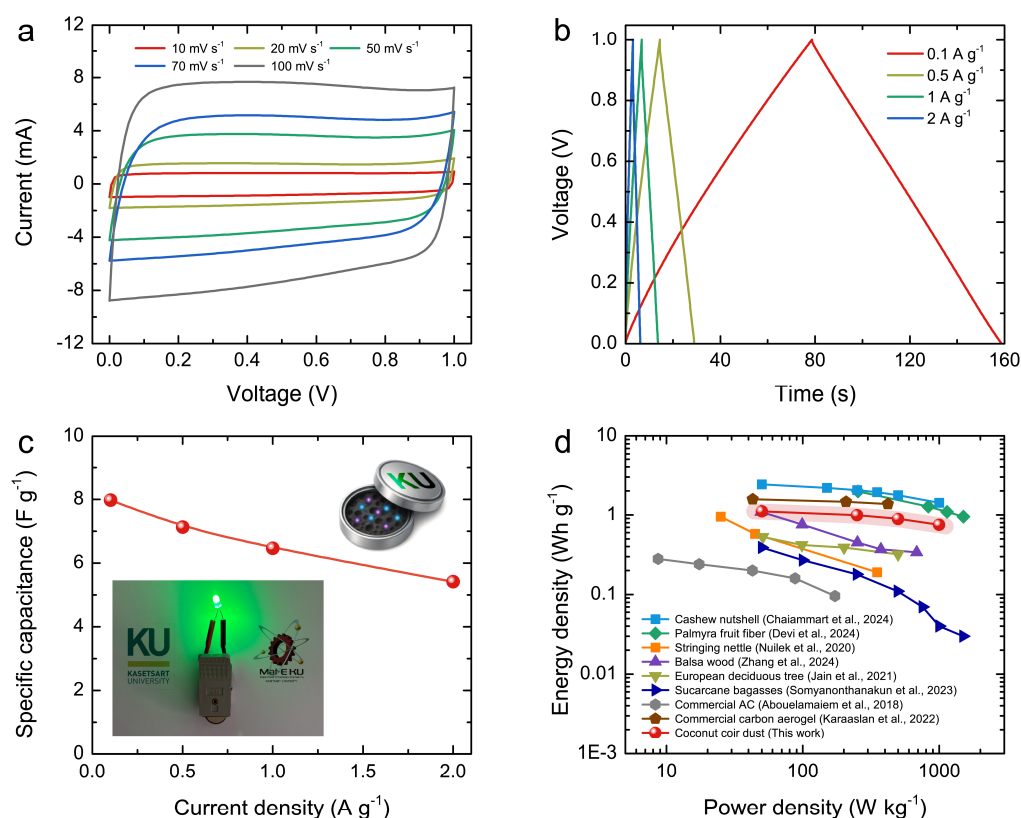


Figure 8. Electrochemical measurements of a symmetric coin-cell supercapacitor device (two-electrode system) using SA-CCD-7 as the electrode materials: (a) CV curves at various scan rates ranging from 10 to 100 mV s⁻¹, (b) GCD curves at various current densities ranging from 0.1 to 2 A g⁻¹, (c) specific capacitance versus current density (the inset shows a green LED powering by two coin-cell supercapacitors), and (d) Ragone plot (energy density versus power density) of our coin-cell supercapacitor device compared with those using biomass-derived ACs as the electrode materials reported in the literature [16,57–63].

The cycling stability of the device was evaluated over 10,000 charge–discharge cycles at 2 A g⁻¹, as shown in Figure 9a. The device exhibited negligible capacitance loss. The first and final GCD curves remained nearly identical, indicating excellent electrochemical re-

versibility and mechanical integrity. Similarly, the GCD curves recorded after 10,000 cycles (Figure 9b) showed minimal distortion compared to the initial scan, confirming excellent long-term durability. Collectively, these results demonstrate that SA-CCD-7 delivered high capacitance, competitive energy and power densities, and exceptional cycling stability, highlighting its potential as a sustainable electrode material for next-generation biomass-derived supercapacitors.

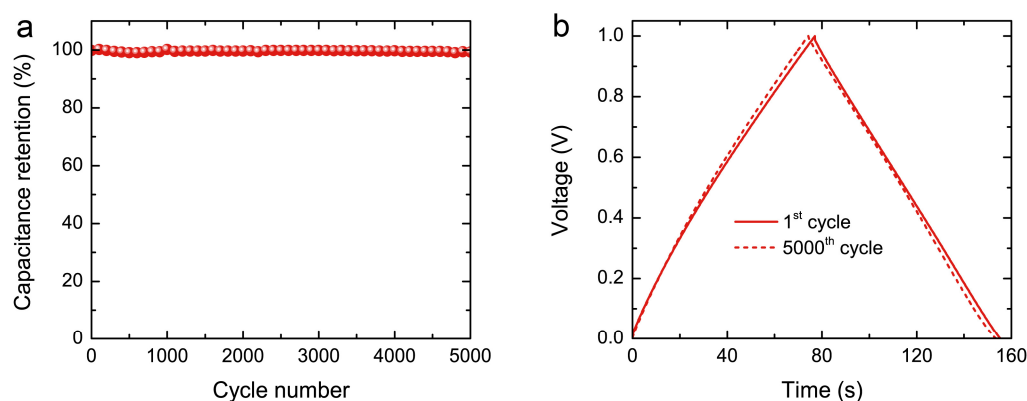


Figure 9. (a) Capacitance retention over 10,000 cycles at 2 A g^{-1} , and (b) GCD curves at 1 A g^{-1} before and after stability test for 10,000 cycles (1st and 10,000th cycles).

4. Conclusions

SA-CCD samples were successfully synthesized from coconut coir dust through carbonization followed by steam activation. The carbonization temperature significantly influenced the resulting pore structure, surface area, and electrochemical performance. The SA-CCD showed a hierarchical micro–mesoporous architecture and well-preserved structural integrity. Among the prepared samples, SA-CCD-7 exhibited the most favorable textural properties, with a specific surface area of $889 \text{ m}^2 \text{ g}^{-1}$ and a total pore volume of $0.715 \text{ cm}^3 \text{ g}^{-1}$. The electrochemical measurements revealed that SA-CCD-7 had the highest specific capacitance of 86 F g^{-1} at 1 A g^{-1} , retaining 81% of its initial capacitance at 20 A g^{-1} , indicating good rate capability. The Trasatti analysis confirmed that charge storage was dominated by EDLC behavior. A symmetric coin-cell supercapacitor assembled with SA-CCD-7 as electrodes delivered an energy density of 1.2 Wh kg^{-1} at a power density of 50 W kg^{-1} , maintaining 0.9 Wh kg^{-1} even at 2500 W kg^{-1} . Moreover, the device exhibited negligible capacitance loss after 10,000 cycles, confirming its excellent long-term stability. This work demonstrates that steam activation of coconut coir dust provides a green, sustainable, and industrially feasible approach for converting agricultural waste into high-performance carbon electrodes. The combination of environmental friendliness, hierarchical porosity, and robust electrochemical performance highlights its potential for next-generation biomass-derived supercapacitors and other sustainable energy-storage technologies.

Supplementary Materials: The following supporting information can be downloaded at <https://www.mdpi.com/article/10.3390/chemistry7060184/s1>: Figure S1: SEM image of raw coconut coir dust; Figure S2: SEM-EDS mapping images and the corresponding spectrum of CCD-7; Figure S3: SEM-EDS mapping images and the corresponding spectrum of SA-CCD-7; Figure S4: CV curves of SA-CCD-5 and SA-CCD-6 at different scan rates from 10 to 100 mV s^{-1} ; Figure S5: GCD curves of SA-CCD-5 and SA-CCD-6 at different current densities from 1 to 20 A g^{-1} ; Table S1: Comparison of specific surface area (S_{BET}) and preparation conditions of ACs synthesized from various coconut-based precursors using chemical and physical activation methods reported in the literature (2020–2025); Table S2: Elemental composition of all CCD and SA-CCD samples measured by XRF analysis. The values were estimated from five measurement areas.

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